

RIC Weekly Note #6

Lecture Content

- Sampling based planning

Literature

- Chapter 7, start until (including) 7.2
We will not go into the math of quasi random sampling
- Section 7.3 (limited focus on this section though)

Suggested Exercises

1. Review the PRM algorithm
2. Review the RRT algorithm
3. Discuss the difference between the Expansive-Spaces Trees (EST) algorithm and the RRT
4. Discuss RRT* and the differences compared to RRT, consider
 - 4.1. Differences in the actual algorithms
 - 4.2. Expected differences in paths planned with RRT vs. RRT*

You can make use of the paper by Karaman & Frazzoli (2010), available on ItsLearning, or also <https://theclassytim.medium.com/robotic-path-planning-rrt-and-rrt-212319121378>